
DSC5001: Statistical Machine Learning I

Week 2: Statistical Inference I: Method of Moments and Maximum Likelihood

Kangning Cui

City University of Hong Kong (Dongguan)

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Course Project Overview

Goal

Apply statistical machine learning methods learned in this course to analyze a real-world data problem.

A successful project should include:

- ▶ A meaningful data question.
- ▶ Appropriate statistical learning methods.
- ▶ Careful evaluation and interpretation.
- ▶ Clear explanation of assumptions and limitations.

Project Organization

- ▶ Group project.
- ▶ Recommended group size: 3–6 students.
- ▶ Each group selects a dataset and defines a statistical learning problem.
- ▶ Every member should contribute to modeling, analysis, and presentation.

Suggested workflow:

1. Identify a research question and dataset.
2. Explore and preprocess the data.
3. Build and compare statistical learning models.
4. Interpret results and discuss limitations.

Possible Project Directions

Healthcare and Biomedical Data

Examples:

- ▶ Disease prediction from medical records
- ▶ Medical image classification
- ▶ Patient risk estimation

Remote Sensing and Geospatial Data

Examples:

- ▶ Land-cover classification
- ▶ Environmental monitoring
- ▶ Crop or vegetation analysis

Other Data Applications

Examples:

- ▶ Financial prediction
- ▶ Text classification
- ▶ Customer behavior analysis

From a Data Question to a Learning Problem

Prediction

Question:

What numerical value do we want to estimate?

Examples:

- ▶ Disease risk score
- ▶ Crop yield estimation
- ▶ Price prediction

Classification

Question:

Which category does an observation belong to?

Examples:

- ▶ Disease / no disease
- ▶ Land-cover type
- ▶ Text category

Structure Discovery

Question:

What hidden patterns exist in the data?

Examples:

- ▶ Patient groups
- ▶ Image clusters
- ▶ Data visualization

How to Choose a Good Project

A good project usually has:

1. **A meaningful question**

What do we want to predict, classify, or understand?

2. **A suitable dataset**

Enough observations, meaningful variables, and reliable labels.

3. **Appropriate methods**

Choose methods based on the problem and data characteristics.

4. **Meaningful evaluation**

Compare models and explain what the results mean.

Learning objectives

1. Distinguish a parameter, an estimator, and an estimate in statistical inference.
2. Construct Method of Moments estimators by matching population and sample moments.
3. Derive and interpret maximum-likelihood estimators for Bernoulli and Gaussian models.
4. Compare estimators using bias, variance, and consistency as sample size increases.

Statistical Inference Roadmap

In statistical inference, we start with an unknown population parameter.

Unknown parameter

We collect data and construct an estimator:

Data $\rightarrow$ Estimator $\rightarrow$ Estimate

Different methods provide different estimators:

Method of Moments



Maximum Likelihood

Finally, we evaluate whether an estimator is reliable:

Bias, Variance, Consistency

From a population model to an estimate

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A Coin-Flipping Example

Suppose we flip a coin three times.
Each flip has two possible outcomes:

Head or Tail.

The observed outcomes are:

Head, Head, Tail.

We want to answer:

What is the probability of observing Head for this coin?

The true probability of Head is unknown. We denote it by

$$p = \mathbb{P}(\text{Head}).$$

Representing the Observed Data

To analyze coin flips mathematically, we represent each outcome by a number.
Define

$$X_i = \begin{cases} 1, & \text{if the } i\text{-th flip is Head,} \\ 0, & \text{if the } i\text{-th flip is Tail.} \end{cases}$$

For the observed sequence

Head, Head, Tail,

the numerical observations are

$$X_1 = 1, \quad X_2 = 1, \quad X_3 = 0.$$

Now the data can be written as

$$(x_1, x_2, x_3) = (1, 1, 0).$$

Parameter, Estimator, and Estimate

In this coin example: $p = \mathbb{P}(\text{Head})$ is the unknown **parameter**.

A rule that uses observations to estimate the parameter is called an **estimator**:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i.$$

For the observed data $(x_1, x_2, x_3) = (1, 1, 0)$, the estimator gives

$$\hat{p} = \frac{1 + 1 + 0}{3} = \frac{2}{3}.$$

This numerical value is called the **estimate**.

- ▶ Parameter: the unknown quantity we want to learn.
- ▶ Estimator: the rule applied to data.
- ▶ Estimate: the value obtained from one dataset.

Repeated Coin Flips

Usually, we do not observe only three flips.
Suppose we flip the same coin n times:

$$X_1, X_2, \dots, X_n.$$

We assume that:

- ▶ Each flip follows the same probability model with the same unknown parameter p .
- ▶ The outcome of one flip does not affect another flip.

These assumptions are summarized as:

$$X_1, \dots, X_n$$

are **independent and identically distributed (i.i.d.)** samples.

Why Independence Matters

For independent observations,

$$p(x_1, \dots, x_n | p) = \prod_{i=1}^n p(x_i | p).$$

The joint probability becomes a product of individual probabilities.
This property will allow us to construct likelihood functions later.

Method of Moments

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The Idea of Method of Moments

Suppose we have a probability model:

$$X \sim f(x; \theta)$$

where θ is an unknown parameter.

The model describes the population, but we only observe a sample:

$$X_1, X_2, \dots, X_n$$

From the observed sample, we calculate summary quantities such as:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Method of Moments estimates θ by making the model-based quantity agree with the corresponding sample quantity.

Population Quantity and Sample Quantity

The probability model gives theoretical population quantities. For example, the population mean is: $E[X]$. The observed data give the sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Method of Moments uses the relationship:

$$E[X] \approx \bar{X}$$

Because $E[X]$ depends on the unknown parameter, we can solve this equation to obtain an estimator.

Main idea

The model describes what should happen in the population. The data show what happened in our sample.

A Simple Example: Bernoulli Model

Consider the coin-flipping model:

$$X \sim \text{Bernoulli}(p)$$

where p is the unknown probability of observing Head.

For a Bernoulli random variable:

$$E[X] = p$$

From observed flips, the sample mean is:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Method of Moments matches:

$$p = \bar{X}$$

Therefore:

$$\hat{p}_{MM} = \bar{X}$$

Why Do We Need More Moments?

A model may contain more than one unknown parameter.

For example:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

has two unknown parameters:

$$\mu, \sigma^2$$

One equation is not enough to determine two unknown values.

Therefore, we need multiple equations from different population quantities.

Main idea

The number of equations should match the number of unknown parameters.

General Method of Moments

A moment describes a population characteristic.

The k -th moment is:

$$E[X^k]$$

The corresponding sample moment is:

$$\frac{1}{n} \sum_{i=1}^n X_i^k$$

Method of Moments matches:

$$E[X^k] = \frac{1}{n} \sum_{i=1}^n X_i^k$$

Using enough moment equations allows us to estimate all unknown parameters.

Why Do We Need Two Moments for a Gaussian Model?

Consider the Gaussian model:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

There are two unknown parameters:

$$\mu, \sigma^2$$

Method of Moments needs enough equations to estimate all unknown parameters.

Therefore, we need two population quantities.

The first one is the population mean:

$$E[X] = \mu$$

The second one should provide information about the spread of the distribution.

Deriving the Second Moment

For a Gaussian model: $X \sim \mathcal{N}(\mu, \sigma^2)$, the variance describes the spread:

$$\text{Var}(X) = E[(X - \mu)^2] = \sigma^2$$

Expanding the squared term:

$$E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2]$$

Using the linearity of expectation:

$$E[X^2] - 2\mu E[X] + \mu^2 = \sigma^2$$

Because: $E[X] = \mu$, we obtain:

$$E[X^2] = \mu^2 + \sigma^2$$

This gives the second moment equation needed for Method of Moments.

Gaussian Method of Moments: Estimate the Mean

Match the first moment

$$E[X] = \mu = \bar{X}$$

Therefore:

$$\hat{\mu}_{MM} = \bar{X}$$

Main idea

The estimated mean makes the model average agree with the observed average.

Gaussian Method of Moments: Estimate the Variance

Match the second moment

$$E[X^2] = \mu^2 + \sigma^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

Substituting:

$$\hat{\mu}_{MM} = \bar{X}$$

we obtain:

$$\hat{\sigma}_{MM}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2$$

Gaussian Method of Moments: Final Result

$$(\hat{\mu}_{MM}, \hat{\sigma}_{MM}^2) = (\bar{X}, \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2)$$

The mean estimate is the sample mean.

The variance estimate is obtained by matching the second population moment with the corresponding sample moment.

Main idea

Method of Moments chooses parameter values that make important characteristics of the model agree with the observed data.

Maximum Likelihood

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Likelihood from Three Coin Flips

Suppose the observed sequence is H, H, T, and let $p = P(\text{Head})$ be the unknown probability of Head. Because the flips are independent, the probability of observing this particular sequence is:

$$P(\text{H,H,T} \mid p) = p^2(1 - p)$$

Now keep the observed data H, H, T fixed and try different candidate values of p .

Candidate p	Probability of H, H, T
0.2	$0.2^2(0.8) = 0.032$
0.5	$0.5^2(0.5) = 0.125$
$\frac{2}{3}$	$(\frac{2}{3})^2(\frac{1}{3}) \approx 0.148$
0.8	$0.8^2(0.2) = 0.128$

Among these candidate values, $p = \frac{2}{3}$ makes the observed data most likely.

Main idea

We change the candidate parameter and ask how well each value explains the observed data.

What Is a Likelihood Function?

Suppose $X_1, \dots, X_n$ are observations generated from a probability model with an unknown parameter θ .

- ▶ X_i : the random variable representing the i -th observation before the data are observed.
- ▶ x_i : the actual observed value of X_i .
- ▶ n : the number of observations in the sample.

Definition

The likelihood function measures how compatible different parameter values are with the observed data.

For observed data $x_1, \dots, x_n$, the likelihood is:

$$L(\theta; x_1, \dots, x_n) = p(x_1, \dots, x_n \mid \theta)$$

If the observations are independent:

$$L(\theta; x_1, \dots, x_n) = \prod_{i=1}^n p(x_i \mid \theta)$$

Probability and Likelihood Use the Same Model Differently

The expression $p(x \mid \theta)$ can be viewed in two different ways.

	Probability model	Likelihood
What is fixed?	Parameter θ	Observed data x
What changes?	Possible data x	Candidate parameter θ
Question	What data might we observe?	Which parameter better explains the observed data?

In maximum likelihood estimation, the observed data have already been collected. We therefore keep $x_1, \dots, x_n$ fixed and compare different values of θ .

Common mistake

Likelihood is not the probability that a parameter value is true.

Why Use the Log-Likelihood?

For independent observations, likelihood multiplies many probability or density terms:

$$L(\theta) = \prod_{i=1}^n p(x_i \mid \theta)$$

Products become inconvenient when n is large.

Because the logarithm is strictly increasing, maximizing $L(\theta)$ is equivalent to maximizing $\log L(\theta)$. We define the **log-likelihood**:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log p(x_i \mid \theta)$$

Maximum Likelihood Estimation

Definition

The maximum likelihood estimator chooses the parameter value that maximizes the likelihood of the observed data.

The estimator is:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta) = \arg \max_{\theta} \ell(\theta)$$

For the coin example, the observed data are H, H, T and:

$$L(p) = p^2(1 - p)$$

Maximum likelihood estimation asks:

Which value of p maximizes $p^2(1 - p)$?

The next step is to solve this optimization problem.

Bernoulli Maximum Likelihood: Build the Likelihood

Assume $X_1, \dots, X_n$ are independent Bernoulli random variables with unknown parameter p .

For each observation, $X_i = 1$ represents Head and $X_i = 0$ represents Tail.

After observing the data, let $x_1, \dots, x_n$ be the observed zeros and ones, and define $m = \sum_{i=1}^n x_i$ as the number of Heads.

Therefore, there are m Heads and $n - m$ Tails.

The likelihood is:

$$L(p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^m (1-p)^{n-m}$$

Taking the logarithm gives:

$$\ell(p) = m \log p + (n - m) \log(1 - p)$$

Bernoulli Maximum Likelihood: Find the Maximum

To find the value of p that maximizes the log-likelihood, differentiate with respect to p :

$$\ell'(p) = \frac{m}{p} - \frac{n-m}{1-p}$$

Set the derivative equal to zero:

$$\frac{m}{p} = \frac{n-m}{1-p}$$

Solving for p gives:

$$m(1-p) = (n-m)p \quad \Rightarrow \quad \hat{p}_{\text{MLE}} = \frac{m}{n}$$

$$\hat{p}_{\text{MLE}} = \frac{m}{n} = \bar{x}$$

The MLE is therefore the observed proportion of Heads.

For H, H, T, $m = 2$ and $n = 3$, so $\hat{p}_{\text{MLE}} = \frac{2}{3}$.

Why Is This a Maximum?

Setting the first derivative to zero gives a stationary point, but we still need to know whether it is a maximum.

For the Bernoulli log-likelihood:

$$\ell''(p) = -\frac{m}{p^2} - \frac{n-m}{(1-p)^2}$$

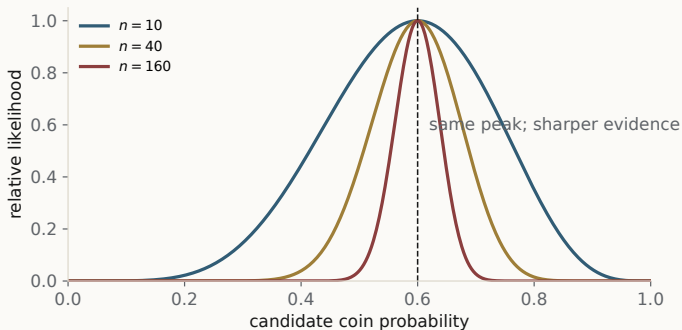
For $0 < p < 1$, this second derivative is negative whenever both Heads and Tails are observed. Therefore, the log-likelihood is concave and the stationary point is a maximum.

Likelihood and Sample Size

Two datasets can produce the same maximum likelihood estimate but provide different amounts of evidence.

Main idea

The location of the maximum gives the estimate. The shape of the likelihood tells us how strongly the data support values near that estimate.



Gaussian Maximum Likelihood: Model Setup

Now assume $X_1, \dots, X_n$ are independent Gaussian random variables with $X_i \sim \mathcal{N}(\mu, \sigma^2)$.

The unknown parameters are the population mean μ and population variance σ^2 .

After observing the sample, the observed values are $x_1, \dots, x_n$.

For one observation, the Gaussian density is:

$$p(x_i | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Because the observations are independent, the likelihood is the product of these n densities.

Gaussian Maximum Likelihood: Log-Likelihood

Multiplying the Gaussian densities and taking the logarithm gives:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

The term $-\frac{n}{2} \log(2\pi)$ does not contain either μ or σ^2 .

Therefore, it cannot change which parameter values maximize the log-likelihood.

For optimization, we can write:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 + C$$

where C is constant with respect to μ and σ^2 .

Gaussian Maximum Likelihood: Estimate the Mean

First, treat σ^2 as fixed and maximize the log-likelihood with respect to μ .

Differentiate:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

Set the derivative equal to zero:

$$\sum_{i=1}^n (x_i - \mu) = 0$$

Therefore:

$$\hat{\mu}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

Main idea

The estimated Gaussian mean is placed at the center of the observed data.

Gaussian Maximum Likelihood: Estimate the Variance

Now substitute $\hat{\mu}_{\text{MLE}} = \bar{x}$ and maximize with respect to σ^2 .

Differentiate:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \bar{x})^2$$

Set the derivative equal to zero and solve for σ^2 :

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Main idea

After estimating the center of the Gaussian distribution, the variance is determined by the average squared distance of the observations from that center.

Maximum Likelihood: What Did We Obtain?

For the Bernoulli model:

$$\hat{p}_{\text{MLE}} = \frac{\text{number of Heads}}{\text{number of flips}}$$

For the Gaussian model:

$$\hat{\mu}_{\text{MLE}} = \bar{x}, \quad \hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

For these examples, Maximum Likelihood and Method of Moments lead to the same estimators. However, the two methods start from different principles:

- ▶ Method of Moments matches population moments with sample moments.
- ▶ Maximum Likelihood chooses parameters that make the observed data most compatible with the model.

Estimator Properties

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Why Do We Study Estimator Properties?

An estimator uses data to estimate an unknown parameter.

However, different samples from the same population usually produce different estimates.

For example, estimating a coin probability from different groups of flips may give:

$$\hat{p}_1, \hat{p}_2, \hat{p}_3, \dots$$

Therefore, we do not only ask:

"What is the estimate?"

We also ask:

- ▶ Is the estimator correct on average?
- ▶ How much does it change across samples?
- ▶ Does it become more accurate with more data?

These questions lead to bias, variance, and consistency.

Repeated Samples and Estimator Behavior

Suppose we repeatedly collect samples from the same population.
Each sample produces an estimate:

$$\hat{\theta}^{(1)}, \hat{\theta}^{(2)}, \dots, \hat{\theta}^{(B)}$$

where θ is the unknown population parameter.

The collection of these estimates describes the behavior of the estimator.

We evaluate an estimator from three perspectives:

1. Accuracy: Is the average estimate close to the true parameter?
2. Stability: Do different samples produce similar estimates?
3. Reliability: Does more data improve the estimate?

Example: Estimating a Coin Probability

Suppose the true probability of Head is:

$$p = 0.5$$

We use the sample proportion:

$$\hat{p} = \frac{\text{number of Heads}}{n}$$

to estimate p .

Now repeat the experiment many times. Each sample gives a different estimate:

$$0.46, \quad 0.52, \quad 0.49, \quad 0.55, \dots$$

The distribution of these estimates describes the behavior of the estimator.

Bias

Definition

Bias measures the systematic difference between the average estimate and the true parameter value.

The bias of an estimator is:

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

where $E[\hat{\theta}]$ represents the average estimate over repeated samples.

An estimator is unbiased if:

$$E[\hat{\theta}] = \theta$$

Main idea

An unbiased estimator may not be correct for every sample, but it is correct on average.

Example: Understanding Bias

Suppose the true parameter is:

$$p = 0.5$$

Consider two estimators from repeated samples. Estimator A produces estimates:

$$0.48, \quad 0.51, \quad 0.49, \quad 0.52$$

The average estimate is close to 0.5.

Estimator B produces estimates:

$$0.60, \quad 0.61, \quad 0.59, \quad 0.62$$

The estimates are consistently above the true value. Estimator B has positive bias.

Main idea

Bias describes whether an estimator is systematically too high or too low.

Variance

Definition

Variance measures how much the estimate changes when we use different samples.

The variance of an estimator is:

$$\text{Var}(\hat{\theta}) = E[(\hat{\theta} - E[\hat{\theta}])^2]$$

A small variance means that repeated samples produce similar estimates.

Main idea

Variance describes the randomness caused by sampling.

Example: Understanding Variance

Suppose the true parameter is:

$$p = 0.5$$

Two estimators have the same average estimate.

Estimator A:

0.49, 0.50, 0.51, 0.50

Estimator B:

0.20, 0.80, 0.35, 0.65

Both may be correct on average. However, Estimator B changes much more across samples. Therefore, Estimator B has higher variance.

Main idea

Variance measures the stability of an estimator.

Bias and Variance Describe Different Errors

Two estimators can have different behaviors.

High bias

- ▶ The estimator is systematically away from the true value.
- ▶ More data may not remove the systematic error.

High variance

- ▶ The estimator changes substantially across samples.
- ▶ A different dataset may give a very different estimate.

A good estimator should control both types of error.

Consistency

Definition

Consistency means that an estimator approaches the true parameter value as the amount of data increases.

For a consistent estimator:

$$\hat{\theta} \rightarrow \theta$$

as the sample size increases.

In other words, with enough observations, the estimator should recover the true population parameter.

Main idea

Consistency describes the long-term behavior of an estimator.

Example: Understanding Consistency

Suppose we estimate a coin probability using:

$$\hat{p} = \frac{\text{number of Heads}}{n}$$

With a small sample:

$$n = 10$$

the estimate may fluctuate.

With a larger sample:

$$n = 10000$$

the estimate usually becomes closer to the true probability.

A consistent estimator becomes more accurate as more data are collected.

Comparing Estimators

When comparing two estimators, no single property is always enough.

An estimator with lower bias may have higher variance.

An estimator with lower variance may have larger systematic error.

Therefore, we should consider:

- ▶ The amount of available data.
- ▶ The type of error that matters.
- ▶ The final goal of the statistical model.

Likelihood in Modern Machine Learning

Maximum likelihood is not only a classical statistical method.

Many modern probabilistic models are trained by maximizing likelihood or minimizing negative log-likelihood. The idea remains the same:

Find parameter values that make the observed data most compatible with the model.

Main idea

Modern models may be complex, but their statistical objectives are often built on classical inference ideas.

Week 2: connect the statistical question, mathematics, and evidence

Questions and discussion